# Particulate Matter Sampling and Analysis

***Reading Material***

***Companion text for the PM sampling and analysis training presentation***

*Prepared from the teaching deck on PM sampling and analysis, with added explanatory material, examples, definitions and study questions.*

## How to use this document

This document is written as reading material for students. Each subsection follows the title of a slide from the presentation, but the text is self-contained: it explains the concepts in complete sentences rather than acting as speaker notes. The aim is to help students understand why particulate matter is measured, how representative samples are collected, how PM mass and composition are determined, and how data quality is evaluated.

## Learning objectives

- Explain why PM10, PM2.5, PM1 and ultrafine particles require different sampling approaches.
- Describe the main components of a PM sampling system, including inlets, flow control, filters and metadata.
- Calculate PM mass concentration from filter mass change and sampled air volume.
- Select suitable filter media and analytical methods for elements, ions, carbonaceous aerosol and organic aerosol.
- Recognize common artifacts, contamination pathways and quality-control checks in PM sampling.
- Use chemical reconstruction as a practical check of mass closure and data consistency.

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# Part A — Background and key concepts

## Particulate Matter Sampling and Analysis

Particulate matter (PM) sampling and analysis is the measurement chain used to determine how much particle pollution is present in air and what that particle pollution is made of. PM cannot be understood from a single measurement alone. The sampling inlet defines which size fraction is collected, the sampler and filter determine what material is retained, and the analytical method determines which chemical information can be obtained.

A useful PM dataset therefore starts with a clear objective. A campaign designed for regulatory mass monitoring may require a reference gravimetric method and 24-hour samples. A campaign designed for chemical composition may require more than one filter type, duplicate samples, field blanks and several laboratory analyses. A campaign designed to study short-term episodes may also require online instruments with hourly or sub-hourly resolution.

In this course, PM analysis is treated as a complete system: sampling design, filter choice, field handling, gravimetric weighing, chemical analysis, quality assurance and reporting all influence the final scientific value of the data.

## Scope of the presentation

The material begins with basic definitions and the reasons PM is monitored. It then moves into practical sampling design: how to select a PM fraction, how to choose a sampler and filter, how to condition and store filters, and how to document field conditions. The second part focuses on the laboratory: mass determination, analytical methods for chemical components, data validation and mass reconstruction.

A central theme is traceability. Every reported concentration should be traceable back to a specific sample, filter ID, sampler, inlet, flow rate, sampling duration, weighing procedure and analytical method. Without this traceability, even a dataset with many samples can be difficult to interpret or compare with other studies.

The presentation also separates sampling and analysis from source apportionment. Source identification requires high-quality chemical and physical data; however, this document focuses on how those data are produced before advanced interpretation begins.

## Why PM measurements matter

PM is one of the most important air pollutants because particles can be inhaled, transported over long distances, scatter or absorb radiation, affect visibility, and participate in cloud processes. Fine particles are especially important for health because they can penetrate deeply into the respiratory system. The chemical composition of PM also matters: particles rich in transition metals, combustion-related carbon or certain organic compounds may have different health relevance from particles dominated by sea salt or mineral dust.

PM measurements are needed for several purposes. Regulatory networks use PM data to assess compliance with air-quality standards. Research campaigns use PM data to understand atmospheric processes, pollution episodes, transport and exposure. Health studies use PM mass and composition to investigate relationships between pollution and health outcomes. Long-term monitoring uses consistent PM measurements to identify whether air quality is improving or deteriorating.

Because PM is a mixture, mass concentration alone is often not enough. Two samples can have the same PM2.5 mass but very different chemical composition and likely different origins. This is why sampling for chemical analysis is a major part of modern aerosol research.

## Current PM benchmarks used in air-quality assessment

Air-quality benchmarks provide context for PM measurements, but they are not a substitute for the measurement method itself. The World Health Organization 2021 global air-quality guidelines recommend

annual mean values of 5 micrograms per cubic metre for PM2.5 and 15 micrograms per cubic metre for PM10, with 24-hour guideline values of 15 and 45 micrograms per cubic metre respectively. These values are health-based guideline levels, not legal limits in every country.

Regulatory standards vary by jurisdiction. The European Union revised its ambient air-quality rules in 2024 and set stricter 2030 standards, including a PM2.5 annual limit value of 10 micrograms per cubic metre. In the United States, the EPA strengthened the primary annual PM2.5 standard to 9 micrograms per cubic metre in 2024, while retaining the 24-hour PM2.5 standard at 35 micrograms per cubic metre. Students should always check the current legal requirements for the country or monitoring programme they are working in.

As guideline and regulatory values decrease, the quality of low-concentration measurements becomes more important. Blank correction, flow accuracy, weighing stability, detection limits and uncertainty estimation are not minor technical details; they can determine whether a reported value is meaningful.

## What are aerosols and particulate matter?

An aerosol is a suspension of solid or liquid particles in a gas. In atmospheric science, the gas is usually air, and the particles are referred to as atmospheric particles or particulate matter. PM can include mechanically generated particles such as dust, combustion particles such as soot and smoke, liquid droplets such as mist, and particles formed by atmospheric reactions.

The terms used for aerosols often describe how they are formed. Dust is usually produced by mechanical disintegration of soil, rock, road material or industrial materials. Smoke is produced by incomplete combustion. Fog and mist are liquid droplets produced by condensation or atomisation. Haze often refers to reduced visibility caused by small particles that scatter light.

Ambient PM is rarely one pure material. It is a mixture of primary particles emitted directly and secondary particles formed in the atmosphere. Its size, composition and physical state can change during transport because of chemical reactions, condensation, evaporation and water uptake.

## PM is characterized by aerodynamic size

PM sampling is based mainly on aerodynamic diameter rather than simple geometric diameter. Aerodynamic diameter describes how a particle behaves in air and is affected by particle size, density and shape. This concept is important because samplers separate particles according to their aerodynamic behaviour, not by looking at them under a microscope.

Common size fractions include TSP, PM10, PM2.5, PM1 and ultrafine particles. TSP, or total suspended particles, includes a broad fraction of airborne particles. PM10 includes particles with aerodynamic diameters smaller than about 10 micrometres. PM2.5 includes particles smaller than about 2.5 micrometres and is often called fine PM. Particles between PM2.5 and PM10 are often called coarse PM. Ultrafine particles are usually smaller than about 100 nanometres and are often more important by number concentration than by mass.

Size determines atmospheric lifetime, deposition in the respiratory tract, transport distance and chemical composition. Coarse particles often settle faster and are enriched in crustal material and sea salt. Fine particles can remain suspended longer and often contain secondary sulfate, nitrate, ammonium, organic aerosol and combustion-related carbon.

## Why size-selective sampling is essential

A PM10 sample, a PM2.5 sample and an ultrafine particle measurement do not represent the same pollutant fraction. PM10 is useful for inhalable particles and includes many coarse components such as dust and sea salt. PM2.5 is more closely associated with fine combustion aerosol and secondary aerosol. Ultrafine particles can be important near fresh emissions but contribute little to total particle mass because each particle is extremely small.

The chosen PM fraction must match the question. If the objective is regulatory PM10 compliance, a PM10 reference or equivalent sampler is needed. If the objective is fine-particle exposure or secondary aerosol chemistry, PM2.5 or PM1 may be more appropriate. If the objective is fresh traffic emissions or nucleation, particle number size distribution may be needed in addition to filter-based mass.

Size-selective inlets must be operated at their specified flow rates. A PM10 inlet used at the wrong flow rate no longer has the intended cut-point, which can bias the sample. For this reason, flow checks are part of the sampling method, not an optional field detail.

> **Key point:** A reported PM value is only as reliable as the sampling definition, field documentation, analytical method and uncertainty estimate behind it.

## Primary and secondary PM

Primary PM is emitted directly into the atmosphere as particles. Examples include mineral dust, sea-salt particles, soot, fly ash, brake wear particles, tyre wear particles and primary organic aerosol. The chemical composition of primary particles can often reflect the source material or combustion process.

Secondary PM forms in the atmosphere from gaseous precursors. Sulfur dioxide can oxidise to sulfate, nitrogen oxides can form nitrate, ammonia can neutralise acidic species to form ammonium salts, and volatile organic compounds can oxidise to form secondary organic aerosol. The formation of secondary PM depends on sunlight, oxidants, temperature, humidity, precursor concentrations and the presence of existing particles.

Sampling captures the particles present at the site and time of measurement. It does not automatically reveal whether a component was emitted directly or formed later. However, chemical composition, size fraction and time patterns can provide important clues about PM formation and transformation.

## What PM is made of

Major PM components include secondary inorganic ions such as sulfate, nitrate and ammonium; carbonaceous material such as organic carbon, elemental carbon and black carbon; mineral dust components such as aluminium, silicon, calcium, titanium and iron; sea-salt components such as sodium and chloride; trace metals such as vanadium, nickel, copper, zinc, arsenic, cadmium and lead; and a large number of organic compounds.

No single analytical method measures all of these components equally well. Ion chromatography is used for water-soluble ions. X-ray fluorescence or digestion followed by ICP-MS can be used for elements. Thermo-optical analysis is used for organic and elemental carbon. Optical methods estimate black carbon or light absorption. Mass spectrometry can provide additional information on organic aerosol composition.

A complete PM analysis plan often requires multiple filters collected in parallel or sequentially. For example, quartz filters are useful for carbon analysis, while PTFE filters are often used for gravimetric mass and elemental analysis. Choosing the filter after the field campaign has started is too late; the analysis plan must be part of the sampling design.

# Part B — PM sampling design and field practice

## End-to-end workflow for PM sampling and analysis

The PM measurement workflow begins before any sample is collected. The first step is to define the objective and data-quality requirements. The second step is to design the sampling approach: site selection, PM fraction, sampler type, filter material, sampling duration, schedule, blanks and duplicates. The third step is field sampling, including flow checks, documentation and safe handling. The fourth step is laboratory analysis. The final step is data validation, uncertainty estimation and reporting.

Students should think of this workflow as a chain. If the chain breaks at any point, the final dataset is weakened. A well-calibrated laboratory method cannot recover information that was lost because the wrong inlet was used, a filter was contaminated, or start and stop times were not recorded.

Planning is especially important for chemical analysis. The selected filter must be compatible with the target analytes, and the sampling duration must provide enough mass for detection without overloading the filter or causing unacceptable artifacts.

## Designing a PM sampling strategy

A sampling strategy should be built around the scientific or regulatory question. For long-term trends, consistent methods and regular sampling are more important than intensive short campaigns. For pollution episodes, high temporal resolution and meteorological information may be more important. For chemical speciation, enough material must be collected for all planned analyses, and the filter media must be compatible with those analyses.

Site selection controls what the sample represents. A regional background site is intended to represent air entering an urban area or the broader region. An urban background site represents population exposure away from direct local sources. A traffic site represents near-road emissions and resuspension. An industrial site may be influenced by stack emissions, fugitive dust or specific metals. Indoor and microenvironment sites require additional attention to occupancy, ventilation and activities.

A good strategy also includes temporal coverage. PM composition changes with season, weekday/weekend patterns, meteorology, heating activity, dust events, biomass burning and photochemistry. A short campaign can be valuable, but its limitations should be stated clearly.

## Selecting the PM fraction and inlet

The inlet defines the particle size fraction. Inlets are designed to transmit or collect particles below a specified aerodynamic cut-point when operated at a specified flow rate. The cut-point is often described by D50, the particle diameter at which 50 percent of particles are transmitted or collected. A PM2.5 inlet, for example, is designed to separate fine particles from larger particles under defined conditions.

Common inlet types include size-selective impactors, cyclones and virtual impactors. Inertial impactors remove larger particles because they cannot follow the air streamlines around an obstacle. Cyclones separate particles using centrifugal forces. Each design has advantages and limitations, including sensitivity to flow rate, particle bounce, grease requirements, loading and maintenance.

The inlet must be clean, correctly assembled and checked during the campaign. Blockages, leaks, insect screens, water intrusion or incorrect flow can change the sampled fraction. For this reason, field logs should include inlet condition and maintenance actions.

## Sampler types and what they are best for

Low-volume samplers are commonly used for routine PM mass and chemical speciation. They are practical, require moderate filter sizes and are often compatible with reference or equivalent methods. High-volume samplers collect more material and are useful when detection limits are challenging or when multiple analyses are needed, but they can be more demanding in terms of power, noise, filter handling and artifacts.

Sequential samplers automatically collect a series of filters and are useful for long campaigns because they reduce manual site visits. Dichotomous samplers can collect fine and coarse fractions separately. Cascade impactors collect size-segregated samples across multiple stages and are useful for size-resolved chemical analysis, although they require careful handling and may be affected by particle bounce.

Online instruments such as beta attenuation monitors, tapered element oscillating microbalances, optical particle counters, online XRF instruments, aerosol chemical speciation monitors and black carbon instruments provide higher time resolution than filters. They are valuable for events and diurnal patterns, but they still need calibration, validation and comparison with reference or filter-based methods.

## Typical filter-sampling train

A filter-sampling train usually includes an inlet, a size selector, a filter holder, a pump, flow-control components, sensors and a data logger. Air enters through the inlet, larger particles are removed by the size selector, and the target fraction is collected on the filter. The pump maintains the required flow, and flow sensors record or control the sampled air volume.

Each component can introduce errors. Leaks reduce the true sampled air volume or allow air to bypass the filter. Poor flow control changes the inlet cut-point and volume calculation. A damaged filter can lose particles or allow bypass. Contaminated filter holders can add mass or chemical species. Temperature and pressure affect air-volume calculations.

Before field deployment, samplers should be leak-checked, flow-checked and cleaned according to a documented procedure. After sampling, the filter should be removed carefully, inspected, labelled and stored under conditions appropriate for the planned analyses.

## Choosing filter media

Filter material should be selected according to the planned analyses. PTFE filters are widely used for gravimetry and elemental analysis because they have low hygroscopicity, good chemical resistance and low background for many elements. Quartz-fibre filters are widely used for organic carbon and elemental carbon analysis and for many organic compounds, but they can adsorb gases and have higher blank levels for some species. Glass-fibre filters can tolerate high temperatures and high loadings but may contain elements that interfere with trace-element analysis.

Nylon filters are useful for some ion measurements, especially nitrate, but can be unsuitable for other analytes. Polycarbonate filters have smooth surfaces and are useful for microscopy and some particle-counting applications. Cellulose and mixed-cellulose ester filters can be useful in specific applications but may have limitations for high-temperature or carbon analyses.

Often the best solution is to collect parallel samples on different filters. For example, one PTFE filter may be used for mass and elements, while one quartz filter is used for EC/OC and organic compounds. The extra effort improves data quality because each filter can be matched to the analysis it supports best.

## Filter handling, conditioning and storage

Filter handling is a major source of uncertainty and contamination. Filters should be handled with clean tweezers, stored in clean labelled containers and protected from dust, moisture, heat, direct sunlight and chemical vapours. Field staff should avoid touching the sampling surface and should document any visible damage, unusual colour, overloading or wetting.

For gravimetric analysis, filters are conditioned before pre-weighing and post-weighing under controlled temperature and relative humidity. Conditioning allows the filter and collected particles to equilibrate with the weighing environment. Without conditioning, filter mass can change because of water uptake or loss rather than because of collected PM.

Storage conditions should match the target analytes. Ions and carbonaceous components may require cool or frozen storage after sampling, depending on method requirements and expected volatility. Organic

analyses can be particularly sensitive to contamination from solvents, plastics and laboratory air. The storage plan should be specified before the campaign begins.

# Part C — Gravimetry, QA/QC and artifacts

## Gravimetric determination of PM mass

Gravimetric analysis determines the collected PM mass from the difference between the post-sampling and pre-sampling filter mass. Although the principle is simple, accurate gravimetry requires a controlled weighing environment, a calibrated microbalance, static-charge control, repeated weighings, balance checks and blank correction.

The mass of collected PM is calculated as: collected mass = post-sampling filter mass minus pre-sampling filter mass, corrected for field blanks when appropriate. Field blanks are filters that travel through handling and transport steps without active sampling. They help estimate contamination or mass changes unrelated to sampled air.

At low PM concentrations, the mass difference can be small. In such cases, balance sensitivity, filter stability, humidity control and blank variability become critical. A reported PM concentration should therefore include or be linked to an uncertainty estimate.

## Air volume and concentration calculations

PM concentration is calculated by dividing the blank-corrected collected mass by the sampled air volume. In its simplest form: $C = m / V$, where C is concentration, m is collected mass and V is sampled air volume. If mass is in micrograms and volume is in cubic metres, concentration is reported in micrograms per cubic metre.

The sampled volume is usually calculated from the average flow rate and sampling duration: $V = Q \times t$. The flow rate must be measured or controlled accurately, and the start and stop times must be recorded. Depending on the method, volume may be reported at ambient conditions or corrected to standard temperature and pressure. The chosen convention must be stated clearly.

A simple example illustrates the calculation. If a PM2.5 filter gains 120 micrograms after blank correction and the sampler collected 24 cubic metres of air, the concentration is 120 / 24 = 5 micrograms per cubic metre. If the flow was wrong or the time was recorded incorrectly, the concentration would be biased even if the weighing was perfect.

> **Formula:** $C = (m_{post} - m_{pre} - m_{blank}) / V$, where V = average flow x sampling time. Always report the PM fraction, averaging period and volume convention.

## QA/QC: what makes PM data defensible?

Quality assurance and quality control make the dataset defensible. Quality assurance is the planned system of procedures: standard operating procedures, training, calibration schedules, documentation and data-validation rules. Quality control is the set of checks applied during and after measurement: blanks, duplicates, flow audits, balance checks, calibration standards, control charts and data flags.

Common QA/QC samples include laboratory blanks, field blanks, travel blanks, replicate filters and certified or reference materials when available. For chemical analysis, calibration standards, continuing calibration checks, matrix spikes and detection-limit assessments may be needed. For online instruments, zero checks, span checks, flow checks and comparison with filter measurements may be appropriate.

QA/QC should not be treated as paperwork. It directly affects whether the data can be used in publications, regulatory assessment or health studies. A dataset without documented blanks, flow checks and uncertainty estimates is much weaker than one with fewer samples but better traceability.

> **Key point:** A reported PM value is only as reliable as the sampling definition, field documentation, analytical method and uncertainty estimate behind it.

## Sampling artifacts and particle losses

Sampling artifacts occur when the collected sample is not an exact representation of the particles in ambient air. Positive artifacts add material to the filter, while negative artifacts remove material from the sample. For example, quartz filters can adsorb organic vapours, producing a positive organic carbon artifact. Ammonium nitrate can evaporate from filters, producing a negative nitrate artifact.

Particle losses can occur in the inlet, sampling lines, impactor stages, filter holder or during handling. Coarse particles may deposit in bends or tubing. Ultrafine particles may be lost by diffusion. Semi-volatile species may evaporate if temperature changes. Hygroscopic particles may gain or lose water depending on relative humidity.

Artifacts cannot always be eliminated, but they can be reduced and documented. Strategies include using appropriate filter materials, denuders, backup filters, short sampling lines, controlled flow, correct storage, rapid transport, and consistent conditioning. The most important step is to recognize which artifacts matter for the target analytes.

## Field metadata: the invisible part of the sample

Field metadata are the information that gives meaning to a filter. At minimum, each sample should have a site name, coordinates or site description, sampler ID, inlet type, PM fraction, filter ID, start time, stop time, flow information, operator name, meteorological notes, sampler status and storage conditions. Additional notes on nearby activities, construction, fires, dust events or instrument alarms can be extremely valuable during interpretation.

Metadata are often called the invisible part of the sample because they are not measured in the laboratory, but without them the laboratory result cannot be interpreted correctly. A filter with excellent chemical analysis but missing sampling duration or flow cannot produce a reliable concentration.

Good metadata systems use pre-prepared field sheets, barcodes or electronic logs. The goal is to reduce free-text ambiguity and make it possible to audit every concentration from final result back to field conditions.

# Part D — Chemical and optical analysis

## Chemical analysis: matching methods to questions

Chemical analysis should be selected according to the questions the study needs to answer. If the focus is mineral dust, elements such as Al, Si, Ca, Ti and Fe are important. If the focus is secondary inorganic aerosol, sulfate, nitrate and ammonium are essential. If the focus is combustion aerosol, EC/OC, black carbon and selected organic markers may be needed. If the focus is toxicity or oxidative potential, transition metals and organic compounds may become especially relevant.

The table below gives a simplified guide. In real campaigns, method choice also depends on detection limits, available sample mass, cost, laboratory capacity, filter compatibility and required time resolution.

| Question | Useful measurement | Typical method or instrument |
|---|---|---|
| How much PM is present? | PM10 or PM2.5 mass | Gravimetric filter method, BAM, TEOM or equivalent monitor |
| Which major ions are present? | SO4, NO3, NH4, Cl, Na, K, Mg, Ca | Ion chromatography after aqueous extraction |
| Which elements and metals are present? | Al, Si, Ca, Ti, Fe, V, Ni, Cu, Zn, Pb and others | XRF, PIXE, ICP-MS or ICP-OES |
| How much carbonaceous aerosol is present? | OC, EC and/or equivalent BC | Thermal-optical carbon analysis; aethalometer or MAAP for optical BC |
| What is the organic composition? | Markers, functional groups or mass spectra | GC-MS, LC-MS, AMS/ACSM, EESI-type methods or other specialized techniques |

**Key point:** Choose the chemical method from the study question and target analytes, then choose the filter media and sampling duration that support that method.

## Elemental analysis of PM filters

Elemental analysis provides information on crustal material, sea salt, metals, industrial emissions, fuel-oil markers, traffic-related wear and other trace components. X-ray fluorescence (XRF) is widely used because it can analyze many elements directly on filters with minimal sample preparation and is generally non-destructive. PIXE is another ion-beam technique used for multi-element analysis. ICP-MS and ICP-OES provide high sensitivity after acid digestion, but the sample is consumed or altered during preparation.

Each method has limitations. XRF detection limits depend on element, filter type, deposit thickness and instrument configuration. ICP methods depend on digestion efficiency, blank contamination and spectral interferences. PIXE requires access to accelerator facilities. For all elemental methods, field blanks and laboratory blanks are essential because trace-metal concentrations can be low.

Elemental data are often used in mass reconstruction and as tracers for PM components. For example, Al, Si, Ca, Ti and Fe are commonly associated with mineral dust, while Na and Cl indicate sea salt when chloride has not been depleted. However, elements can have multiple sources, so interpretation should be cautious.

## Water-soluble ions by ion chromatography

Ion chromatography is commonly used to measure water-soluble ions such as sulfate, nitrate, ammonium, chloride, sodium, potassium, magnesium and calcium. These ions are central to PM composition because they represent secondary inorganic aerosol, sea salt and some contributions from crustal or combustion sources.

The typical procedure is extraction of a portion of the filter in ultrapure water, followed by analysis of anions and cations. Quality control includes extraction blanks, calibration standards, continuing calibration checks, replicate extracts and ionic balance checks where appropriate. Storage and extraction conditions matter because some ions are semi-volatile or chemically reactive.

Nitrate and ammonium are particularly sensitive to sampling artifacts because ammonium nitrate can partition between particle and gas phases. Chloride can also be depleted from sea salt through reaction

with nitric or sulfuric acid. Students should remember that measured ion concentrations reflect both atmospheric composition and sampling conditions.

## Carbonaceous aerosol: OC, EC and BC

Carbonaceous aerosol is a major component of fine PM in many environments. Organic carbon (OC) represents carbon in organic compounds. Elemental carbon (EC) is a refractory carbon fraction commonly associated with combustion and is measured by thermal-optical methods. Black carbon (BC) is an optical quantity related to light absorption by carbonaceous particles. OC, EC and BC are related but not identical.

Thermal-optical analysis heats a filter sample in controlled atmospheres and measures evolved carbon. The optical correction accounts for pyrolyzed carbon formed during heating. Different temperature protocols can produce different OC/EC splits, so the protocol must be reported. Quartz filters are commonly used for OC/EC, but they require careful blank correction because organic vapours can adsorb to quartz.

Black carbon is often measured with optical instruments such as aethalometers or MAAP-type instruments. These instruments provide high-time-resolution information on combustion aerosol but require assumptions about optical properties and corrections for filter loading and scattering effects.

## Organic aerosol and molecular-level analysis

Organic aerosol is chemically complex. It includes primary organic aerosol emitted directly from sources and secondary organic aerosol formed by oxidation of volatile organic compounds. The organic fraction can include thousands of compounds with different volatility, polarity and reactivity. This complexity makes organic aerosol one of the most challenging PM components to measure and interpret.

Bulk OC analysis gives the amount of organic carbon but not the identity of the compounds. Molecular-marker analysis targets specific compounds such as levoglucosan, hopanes, polycyclic aromatic hydrocarbons or biogenic markers. Mass spectrometric methods can provide broader information on organic aerosol composition, oxidation state and temporal variability.

Sampling artifacts are especially important for organics. Semi-volatile organic compounds can evaporate from particles, while gas-phase organics can adsorb to filters. Filter choice, denuders, backup filters, storage temperature and extraction procedures should be selected according to the target organic measurements.

## Optical analysis: black carbon and light absorption

Optical analysis measures how particles absorb or attenuate light. This is useful because black carbon and related light-absorbing particles influence climate and are strong indicators of combustion. Instruments such as aethalometers collect particles on a filter tape and track the change in light attenuation over time.

Optical measurements provide excellent time resolution, but they are operationally defined. The reported value may be equivalent black carbon rather than a direct mass of pure black carbon. The conversion from light absorption to mass depends on an assumed mass absorption cross-section. Filter loading, multiple scattering, particle mixing state and brown carbon can influence the result.

For reporting, it is important to state the instrument type, wavelength, correction algorithm and whether the result is absorption coefficient, equivalent black carbon or another optical quantity. Comparing optical BC values across studies requires attention to these methodological details.

## Online and high-time-resolution PM composition

Filter samples often represent 24-hour averages, which can hide short-term events such as traffic peaks, cooking emissions, fireworks, dust gusts or rapid changes in boundary-layer height. Online instruments provide time resolution from hours to minutes or seconds and can reveal diurnal patterns, emission events and atmospheric processes.

Examples include online XRF instruments for elements, aerosol chemical speciation monitors for non-refractory submicron sulfate, nitrate, ammonium, chloride and organic aerosol, aethalometers for equivalent black carbon, and particle counters or mobility spectrometers for number size distributions. These instruments complement filter sampling rather than replacing it completely.

High-time-resolution instruments require their own QA/QC: calibration, flow checks, time synchronization, data-flagging rules, maintenance logs and comparisons with independent measurements. A high-resolution dataset is powerful only if the time stamps and instrument performance are reliable.

## Sample preparation: keep the analyte in mind

Sample preparation is the bridge between field collection and laboratory measurement. The preparation method must preserve the analyte and avoid introducing contamination. Ion analysis usually uses aqueous extraction. Elemental analysis may use direct filter measurement or acid digestion. Organic analysis may require solvent extraction or thermal treatment. EC/OC analysis usually uses a punched section of a quartz filter.

The same filter cannot always be used for every analysis. Acid digestion destroys the filter deposit for other analyses. Solvent extraction can remove organic compounds but may not be compatible with subsequent elemental or carbon analysis. Cutting a filter into portions requires care to avoid contamination and to ensure that each punch is representative of the deposit.

Good preparation practice includes clean work areas, documented tools, blanks, recovery checks, replicate preparations and chain-of-custody records. Students should learn to ask: What might be lost during preparation? What might be added? How do we know?

# Part E — Data validation, reconstruction and reporting

## Data processing and validation

Raw outputs become scientific data only after processing and validation. For filter samples, processing includes flow validation, sampling-time verification, blank correction, mass calculation, concentration calculation and uncertainty estimation. For chemical data, it includes calibration checks, blank correction, detection-limit treatment, recovery assessment and consistency checks.

Outliers should not be removed automatically. They may represent real events such as dust storms, fireworks, biomass-burning episodes or sampler contamination. Field notes, meteorology, neighbouring sites and instrument diagnostics should be used to decide whether a value is valid, invalid or valid but flagged.

A transparent dataset keeps both the value and the data flag. For example, a sample might be reported as valid, suspect, below detection limit, flow failure, overloaded filter or contaminated blank. Flags allow later users to apply their own selection criteria without losing information.

## Chemical reconstruction and mass closure

Chemical reconstruction, also called mass reconstruction, estimates how much of the measured PM mass can be explained by measured chemical components. It is a useful quality-control and interpretation tool. A simple reconstruction may include secondary inorganic aerosol, organic matter, elemental carbon, mineral dust, sea salt, trace elements and unmeasured or particle-bound water.

A general expression is: reconstructed PM = secondary inorganic ions + organic matter + elemental carbon + mineral dust + sea salt + trace elements + other or unmeasured mass. Secondary inorganic ions are often represented by sulfate, nitrate and ammonium. Organic matter is estimated from organic carbon using an organic-mass-to-organic-carbon factor. Mineral dust is estimated from crustal elements using oxide factors. Sea salt is estimated from sodium and chloride, depending on the available data and possible chloride depletion.

Mass closure compares reconstructed PM with gravimetric PM. Good closure does not prove that every component is perfectly measured, but poor closure indicates that something requires investigation: missing species, incorrect assumptions, blank problems, volatile losses, water content, analytical bias or inconsistent size fractions.

> **Common reconstruction form:** PM_reconstructed = SIA + OM + EC + mineral dust + sea salt + trace elements + other/unmeasured material.

## Chemical reconstruction: assumptions and pitfalls

Chemical reconstruction is useful precisely because it is simple, but its simplicity depends on assumptions. The factor converting OC to organic matter can vary from about fresh hydrocarbon-like aerosol to more oxidised secondary organic aerosol. A single constant may be convenient but not always accurate. Similarly, estimating sulfate from sulfur assumes that sulfur is present as sulfate, which may not always be true.

Mineral dust reconstruction often uses oxide factors for elements such as Al, Si, Ca, Fe and Ti. This assumes crustal composition and full oxidation states. Real dust composition can vary by region, and elements such as Fe or Ca can also have non-crustal sources. Sea-salt reconstruction can be biased if chloride is depleted by reaction with acids.

The best practice is to state the equations and assumptions, test sensitivity where possible, and use reconstruction as a consistency check rather than as an exact source apportionment result. Reconstruction can tell students whether the measured chemical components are broadly consistent with the mass, but it does not by itself identify all emission sources.

## Reporting PM sampling and analysis results

A complete PM report should allow another person to understand exactly how the measurements were produced. It should describe the site, time period, PM fraction, inlet, sampler, flow rate, filter type, sampling duration, conditioning procedure, weighing procedure, analytical methods, calibration, blanks, detection limits, uncertainty estimates and data-validation rules.

Results should be reported with clear units and averaging periods. A concentration in micrograms per cubic metre is incomplete unless the PM fraction and averaging time are also given. Chemical components should specify whether values are reported as mass of the ion, mass of the element, mass of carbon, equivalent black carbon or another operational quantity.

Transparency is essential for comparison. Differences between two datasets may reflect real atmospheric differences, but they may also reflect different inlets, filters, conditioning protocols, analytical methods or data-treatment rules. Good reporting helps separate these possibilities.

> **Key point:** A reported PM value is only as reliable as the sampling definition, field documentation, analytical method and uncertainty estimate behind it.

## Practical checklist for a PM campaign

Before a PM campaign begins, the team should confirm the objective, site permissions, power supply, sampler maintenance, calibrated flow devices, filters, labels, field sheets, blanks, transport containers, storage conditions and safety procedures. Laboratory capacity and analytical schedule should also be confirmed before samples are collected.

During the campaign, operators should record start and stop times, flows, sampler status, filter condition, weather, local activities, maintenance, alarms and any unusual events. Field blanks and duplicates should be collected according to the QA/QC plan, not added only after problems appear.

After the campaign, filters should be transported and stored correctly, data should be checked against field logs, invalid samples should be documented, and analytical results should be reviewed for consistency. The checklist is a simple but powerful tool for preventing avoidable data-quality problems.

## Take-home messages

PM sampling and analysis are not separate activities. They form a single measurement system. The inlet defines the size fraction, the flow defines the sampled volume, the filter collects the particles, the handling preserves or alters the sample, the laboratory method measures selected components, and the validation process determines what can be reported with confidence.

Good PM data require planning, traceability and uncertainty awareness. Students should be able to explain not only what value was measured, but also how it was obtained and what limitations apply. This is especially important for low concentrations, trace components and semi-volatile species.

The final message is practical: choose the method according to the question, document every step, use appropriate blanks and QA/QC, and report assumptions clearly. Reliable PM data are the foundation for exposure studies, air-quality assessment, chemical interpretation and later source-related analysis.

# Glossary of key terms

**Aerodynamic diameter:** The diameter of a unit-density sphere with the same aerodynamic behaviour as the particle of interest.

**Blank correction:** Correction using filters or samples that undergo handling steps without active sampling, used to estimate contamination or non-sampling mass changes.

**Cut-point / D50:** The particle aerodynamic diameter at which a size-selective inlet transmits or collects 50 percent of particles.

**Field blank:** A filter taken to the field and handled like a sample but not used for active air sampling.

**Gravimetry:** Determination of PM mass by weighing a filter before and after sampling under controlled conditions.

**Mass closure:** Comparison between measured gravimetric PM mass and reconstructed PM mass from chemical components.

**PM10:** Particles with aerodynamic diameter below about 10 micrometres, according to the sampling convention of the inlet.

**PM2.5:** Particles with aerodynamic diameter below about 2.5 micrometres, commonly referred to as fine particulate matter.

**Secondary aerosol:** Particle material formed in the atmosphere from gas-phase precursors through chemical reactions and partitioning.

**Uncertainty:** Quantitative estimate of the range within which the true value is expected to lie, considering known sources of measurement error.

## Review questions for students

1. Why is aerodynamic diameter more useful than geometric diameter for PM sampling?
2. What can go wrong if a PM2.5 inlet is operated at an incorrect flow rate?
3. Why might a campaign collect both PTFE and quartz filters in parallel?
4. Explain the difference between OC, EC and BC.
5. What is the purpose of a field blank? Give two examples of problems it can reveal.
6. A filter gains 180 micrograms after blank correction and the sampled air volume is 30 cubic metres. What is the PM concentration?
7. Why can nitrate and some organic compounds be affected by sampling artifacts?
8. What information should be included in field metadata for a PM sample?
9. How can chemical reconstruction be used to check data consistency?
10. Why is chemical reconstruction not the same as source apportionment?

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